

PROJECT 5 — BI DASHBOARD MINI PROJECT

Opportunity & Risk Mapping

Healthcare (SME, 51–250 employees) | AI Implementation Analysis

Prepared for: Chleo, CEO | Date: March 2026 | Confidential

Purpose of This Document

This document maps the specific AI opportunities and risks for Chleo's healthcare SME. It is structured to inform a prioritisation conversation: which opportunities offer the highest return relative to implementation effort, and which risks require mitigation before or during implementation. All assessments are calibrated to the SME context (51–250 employees, limited dedicated IT/data resources, compliance-sensitive environment).

1. Opportunity Map

Opportunities are rated across four dimensions: Feasibility (for SME), Transparency (can the AI explain itself?), ROI Speed (time to measurable return), and Compliance Fit (how well it avoids PHI exposure).

Use Case	Feasibility	Transparency	ROI Speed	Compliance Fit
No-show Prediction	High	High	Fast (1–3 mo)	Excellent (de-id)
Appointment Scheduling AI	Medium	High	Medium (3–6 mo)	Excellent
Clinical Note Summarisation	Medium	Medium	Medium (3–6 mo)	Moderate (PHI)
Patient Intake Automation	High	High	Fast (1–3 mo)	High
RAG Chatbot for Patient FAQs	High	High	Medium (3–6 mo)	High
Billing / Revenue Cycle AI	Low–Med	Medium	Slow (6–12 mo)	Moderate (PHI)
Diagnostic Image AI	Low	Low–Med	Slow (12+ mo)	Low (heavy PHI)

2. Top 3 Prioritised Opportunities

The following three use cases are recommended as the initial scope for Chleo's organisation, selected on the basis of high feasibility, strong explainability, fast ROI, and minimal PHI exposure.

OPPORTUNITY 1 — No-Show Prediction Train a simple ML model on historical appointment data (age, chronic conditions, SMS reminders, lead time) to flag patients at high no-show risk. Staff receive a daily prioritised callback list. Measurable impact within 4–8 weeks. Dataset: 110k appointments (Kaggle). No PHI required for the model — only de-identified scheduling metadata.	OPPORTUNITY 2 — Patient Intake Email Automation n8n workflow monitors the intake inbox, extracts patient request type using an LLM classifier, and routes to the correct department or auto-generates an acknowledgement. Reduces manual triage time by ~60%. Fully auditable — every classification is logged. LangSmith monitors accuracy in real time.
OPPORTUNITY 3 — Clinical Note Summarisation (Monitored) A prompted LLM (Claude/GPT-4) receives a clinician's dictated note and returns a structured SOAP summary. The model never stores PHI — it processes and discards. LangSmith evaluation tracks summary accuracy, flagging hallucinations. This directly addresses Chleo's transparency concern by making AI decisions visible and reviewable.	FUTURE — RAG Patient FAQ Chatbot Phase 2 opportunity: a chatbot trained on the organisation's own policy documents, FAQs, and treatment guides. Answers are grounded in verified internal content (no hallucination of clinical advice). Low PHI exposure. Strong patient engagement ROI.

3. Risk Register

Each risk is assessed on Likelihood (L) and Impact (I) on a 1–3 scale, producing a Risk Score (L x I, max 9). Mitigation strategies are provided for each.

Risk	L	I	Score	Mitigation
PHI exposure in AI pipeline	2	3	6	Use de-identified or synthetic data for all model training. Require BAA from any AI vendor touching real patient data. Audit logs on all data access.
HIPAA / GDPR non-compliance	2	3	6	Engage a HIPAA compliance consultant before go-live. Embed privacy officer review into AI procurement. Monitor HHS OCR guidance updates.
Shadow AI adoption by staff	3	2	6	Implement an approved AI tool list. Provide staff training. Create a fast-track approval process so governed AI is always the easier choice.
Model bias (demographic fairness)	2	3	6	Audit model outputs across patient subgroups (age, gender, chronic condition status) before deployment. Document methodology for regulators.
AI hallucination in clinical notes	2	3	6	Every LLM output requires clinician review before entering the EHR. LangSmith monitoring tracks and flags outputs deviating from expected structure.
Staff resistance / low adoption	3	2	6	Identify clinical champion early. Run 2-week piloted sprint with opt-in staff. Gather structured feedback. Do not force adoption — demonstrate value first.

Vendor lock-in	2	2	4	Prefer open APIs and data-portable solutions. Avoid proprietary AI bundled in EHR contracts without exit clause audit.
Budget overrun	2	2	4	Use free-tier APIs (Anthropic, OpenAI) for PoC. Define hard budget ceiling per sprint. Review actual vs. estimated token costs monthly.
Integration failure with EHR	2	2	4	Run integration as a separate spike sprint. Prefer HL7/FHIR-compatible data exchange. Validate with sandbox before production.

4. Transparency Framework — Addressing Chleo's Core Concern

Chleo's primary fear is that AI is a black box. The following framework demonstrates how every AI component in this project can be made observable and auditable:

Monitored Inputs Every prompt sent to the LLM is logged in LangSmith with timestamp, user ID, and session context. Nothing enters the model unrecorded.	Auditable Outputs Every LLM response is logged alongside the input. Outputs can be reviewed, rated, and flagged by staff or the compliance officer at any time.
Human-in-the-Loop For clinical use cases, no AI output is actioned without human review. AI is a decision-support tool, not a decision-making tool. This is hardcoded into the workflow.	Evaluation Datasets LangSmith evaluation runs the AI against a benchmark dataset weekly, scoring accuracy, hallucination rate, and response format compliance. Results are visible in the dashboard.
Explainable Models The no-show prediction model uses a gradient boosted tree — a model class where feature importance is directly interpretable. Staff can see exactly which factors drove each prediction.	Compliance Audit Trail n8n logs every workflow execution with input, output, and timestamp. Combined with LangSmith, this creates a full end-to-end audit trail satisfying both HIPAA logging requirements and GDPR accountability.

The full monitoring and audit capability will be demonstrated live in the meeting using LangSmith's dashboard — showing real AI inputs, outputs, and evaluation scores in production.